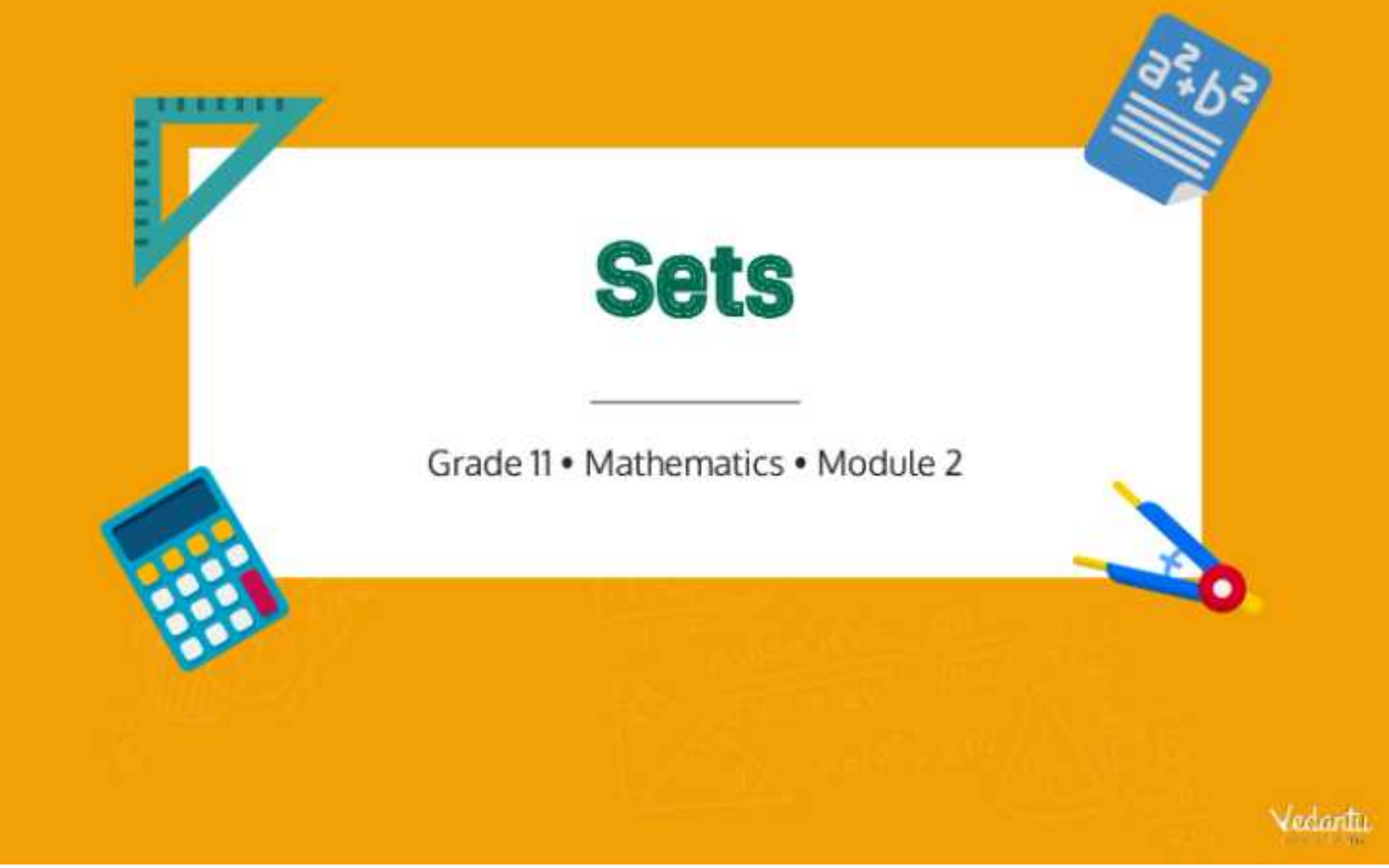
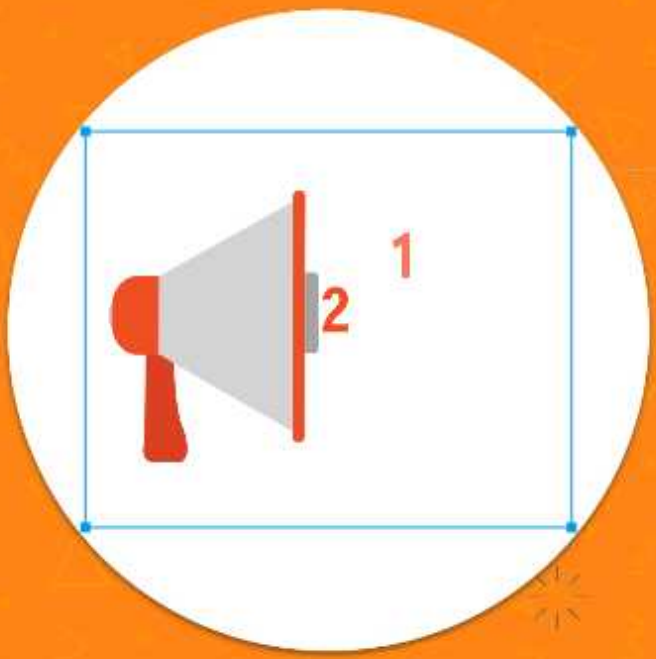

Sets
Grade 11 • Mathematics • Module 2



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Quiz.



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Quiz.

Chat Off

Q. If $A=\{1,2,3,6\}, B=\{1,3,5\}$ and $U=\{1,2,3,4,5,6,7\}$, then which of the following is false?

A $A \cup B=\{1,2,3,5,6\}$

B $A \cap B=\{1,3\}$

C $A / B=\{2,6\}$

D $B / A=\{5,6\}$



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Quiz.

Chat On

Q. If $A=\{1,2,3,6\}, B=\{1,3,5\}$ and $U=\{1,2,3,4,5,6,7\}$, then which of the following is false?

A $A \cup B=\{1,2,3,5,6\}$

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D $B / A=\{5,6\}$



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Quiz.

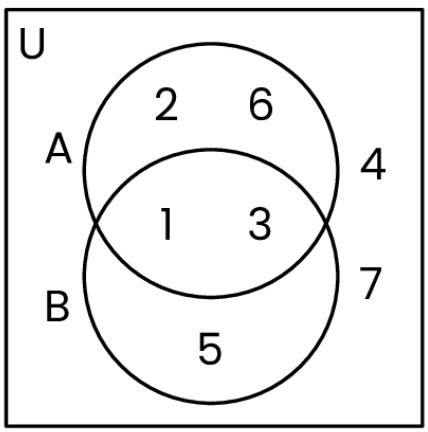
Q. If $A=\{1,2,3,6\}, B=\{1,3,5\}$ and $U=\{1,2,3,4,5,6,7\}$,then which of the following is false?

Solution

$A=\{1,2,3,6\}$ $B=\{1,3,5\}$ $U=\{1,2,3,4,5,6,7\}$

$A\cup B=\{1,2,3,5,6\}$ $A\cap B=\{1,3\}$

$A/B=\{2,6\}$ $B/A=\{5\}$



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Contents

1 Definition and Description of Sets

2 Types of Sets

3 Operations on Sets

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Contents

1 Definition and Description of Sets

2 Types of Sets

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1.6 Cardinality based problems

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1.6 Cardinality based problems

Result

(i) $n(A \cup B) = n(A) + n(B) - n(A \cap B)$

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1.6 Cardinality based problems

Remark

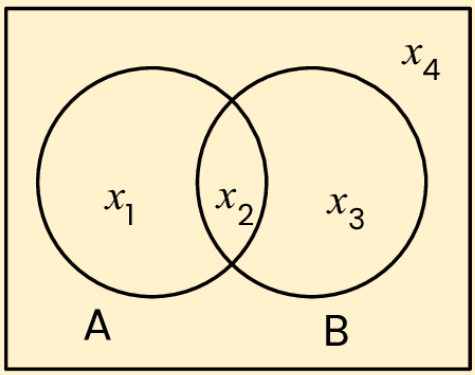
(a) $n(A) = x_1 + x_2$

(b) $n(\text{only } A) = x_1$

(c) $n(B) = x_2 + x_3$

(d) $n(\text{only } B) = x_3$

(e) $n(\text{none of } A \text{ or } B) = x_4$



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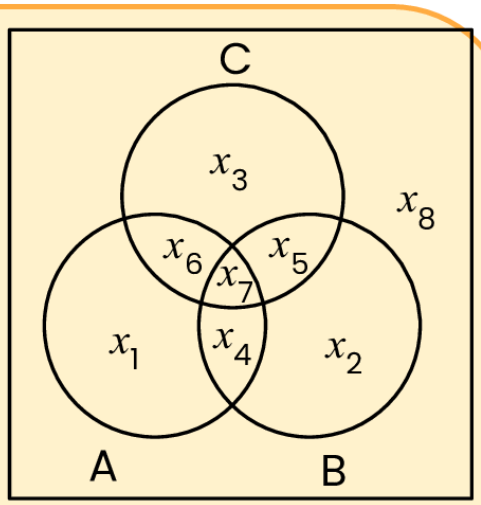
1.6 Cardinality based problems

Remark

(a) $n(A) = x_1 + x_4 + x_6 + x_7$

(b) $n(\text{only } A) = x_1$

(c) $n(\text{only } A \text{ and } B) = x_1 + x_4 + x_2$



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Let's Solve



Let's Solve

Ex. In a school, every teacher teaches either maths or physics. If 12 teach maths, 20 teach physics, while 4 teach both, then find

(a) the number of teachers.

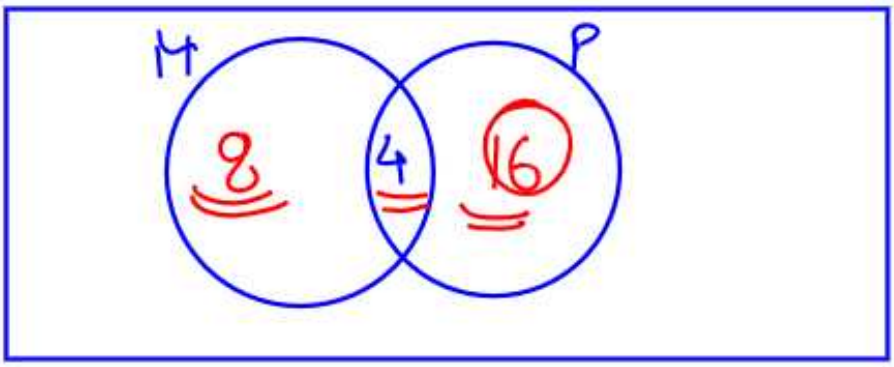
(b) the number of teachers who teach physics only.

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Let's Solve

Ex. In a school, every teacher teaches either maths or physics. If 12 teach maths, 20 teach physics, while 4 teach both, then find

(a) the number of teachers.



$8 + 4 + 16 = 28$

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Let's Solve

Ex. In a school, every teacher teaches either maths or physics. If 12 teach maths, 20 teach physics, while 4 teach both, then find

(a) the number of teachers.

Solution

Let A be the set of teacher who teaches Maths

B be the set of teacher who teaches Physics

Then $n(A)=12, n(B)=20$

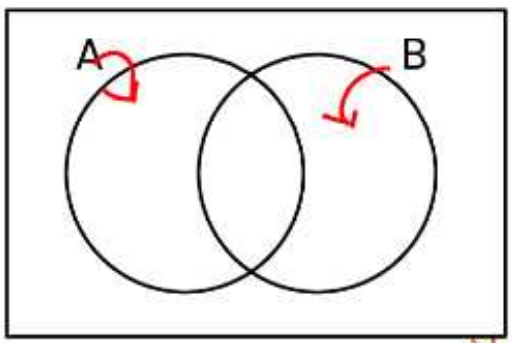
$n(A\cap B)=4$

∴ Total number of teachers

$=n(A\cup B)=n(A)+n(B)-n(A\cap B)$

$=12+20-4$

$=28$



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Let's Solve

Ex. In a school, every teacher teaches either maths or physics. If 12 teach maths, 20 teach physics, while 4 teach both, then find

(b) the number of teachers who teach physics only.

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Let's Solve

Ex. In a school, every teacher teaches either maths or physics. If 12 teach maths, 20 teach physics, while 4 teach both, then find

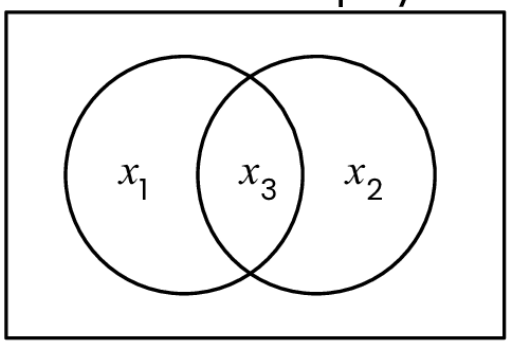
(b) the number of teachers who teach physics only.

Solution

Let $x_1 =$ num of teachers who teach physics only

Let $x_2 =$ num of teachers who teach maths only

Let $x_3 =$ num of teachers who teach both maths and physics



According to the question,

$x_1 + x_3 = 20$ and $x_2 + x_3 = 12$ and $x_3 = 4$

Solving, we get $x_1 = 20 - 4 = 16$

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$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

$$n(A \cup B \cup C) = n(A) + n(B) + n(C) -$$

$$n(A \cap B) - n(B \cap C) - n(C \cap A) +$$

$$n(A \cap B \cap C)$$

Quiz.

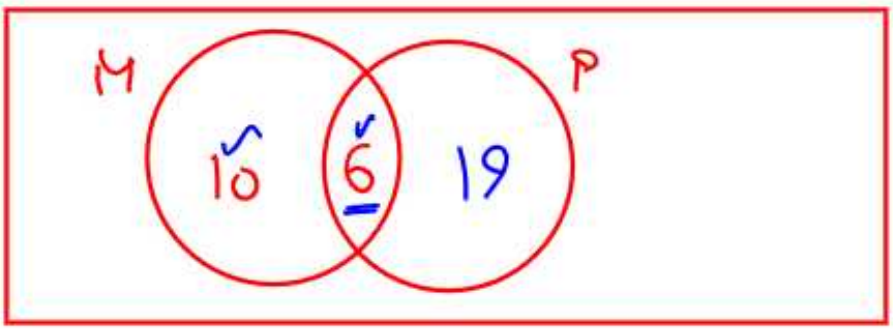


Quiz.

Chat Off

Q. In a school, every teacher teaches either maths or physics. If number of teachers who teach physics is 25, while the number of teachers who teach only maths is 10, and the number of teachers who teach both is 6, then 16 teachers teach maths and in total, there are 35 teachers.

A 16, 32 **B** 16, 35 **C** 19, 32 **D** 19, 35



$10+6+19=35$

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Quiz.

Chat On

Q. In a school, every teacher teaches either maths or physics. If number of teachers who teach physics is 25, while the number of teachers who teach only maths is 10, and the number of teachers who teach both is 6, then _____ teachers teach maths and in total, there are _____ teachers.

A 16, 32

B 16, 35

C 19, 32

D 19, 35

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Quiz.

Q. In a school, every teacher teaches either maths or physics. If number of teachers who teach physics is 25, while the number of teachers who teach only maths is 10, and the number of teachers who teach both is 6, then _____ teachers teach maths and in total, there are _____ teachers.

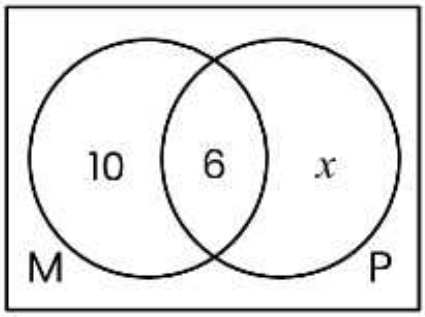
Solution

$n(M) = 10 + 6 = 16$

According to the question, $n(P) = 25$

$So,\ x + 6 = 25 \Rightarrow x = 19$

Total num of teachers $= 10 + 6 + 19 = 35$



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1.6 Cardinality based problems

Observation.

In a school, if 12 teachers teach maths and 20 teachers teach physics, and if total number of teachers is 30, then try to observe that minimum value of $n(M \cap P)$ is 2.

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1.6 Cardinality based problems

Remark

(a) If $n(A)+n(B) \geq n(U)$, then minimum value of $n(A \cap B)$ is given by $n(A)+n(B)-n(U)$

(b) If $n(A)+n(B)<n(U)$, then minimum $n(A \cap B)=0$

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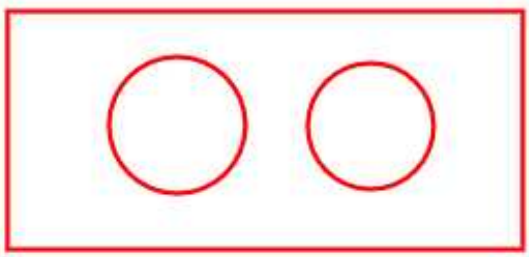
$n(A \cap B)_{min} = ?$

i) $n(A) + n(B) \geq n(U)$

$n(A \cap B)_{min} = n(A) + n(B) - n(U)$

ii) $n(A) + n(B) < n(U)$

$n(A \cap B)_{min} = 0$



Let's Solve



Let's Solve

Ex. If $n(A)=4$ and $n(B)=6$, then find maximum and minimum possible values of $n(A\cap B)$ if

(a) $n(U)=9$

(b) $n(U)=12$

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Let's Solve

Ex. If $n(A)=4$ and $n(B)=6$, then find maximum and minimum possible values of $n(A\cap B)$ if

(a) $n(U)=9$

$n(A)+n(B)=10 > n(U)$

$n(A\cap B)_{min}=10-9=1$

$n(A\cap B)_{max}=4=n(A)$

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Let's Solve

Ex. If $n(A)=4$ and $n(B)=6$, then find maximum and minimum possible values of $n(A\cap B)$ if

(a) $n(U)=9$

Solution

Given, $n(A)=4$ and $n(B)=6$

Here, maximum possible value of $n(A\cap B)=4$

Now, as $n(U)=9$ and $n(A)+n(B)=10$

Thus, minimum value of $n(A\cap B)=1$

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Let's Solve

Ex. If $n(A)=4$ and $n(B)=6$, then find maximum and minimum possible values of $n(A\cap B)$ if

(b) $n(U)=12$

$n(A)+n(B)=10 < n(U)$

$\min n(A\cap B)=0$

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Let's Solve

Ex. If $n(A)=4$ and $n(B)=6$, then find maximum and minimum possible values of $n(A\cap B)$ if

(b) $n(U)=12$

Solution

Given, $n(A)=4$ and $n(B)=6$

Here, maximum possible value of $n(A\cap B)=4$

Now, as $n(U)=12$ and $n(A)+n(B)=10$

Thus, minimum value of $n(A\cap B)=0$

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Quiz.



Quiz.

Chat Off

Q. If $n(A)=4, n(B)=6$ and $n(U)=7$, then find the minimum possible value of $n(A \cap B)$.

A 2 **B** 3 **C** 4 **D** 0

$n(A)+n(B) = 10 \geq n(U)$

$n(A \cap B)_{min.} = 10-7 = 3$



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Quiz.

Chat On

Q. If $n(A)=4, n(B)=6$ and $n(U)=7$, then find the minimum possible value of $n(A \cap B)$.

A 2

B 3

C 4

D 0



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Quiz.

Q. If $n(A)=4, n(B)=6$ and $n(U)=7$, then find the minimum possible value of $n(A \cap B)$.

Solution

Given, $n(A)=4, n(B)=6, n(U)=7.$

Since $n(A)+n(B) \geq n(U)$

$$\begin{aligned} So, \text{minimum } n(A \cup B) &= n(A)+n(B)-n(U) \\ &= 4+6-7 \\ &= 3 \end{aligned}$$

Thus, minimum value of $n(A \cap B)$ is 3

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Let's Solve

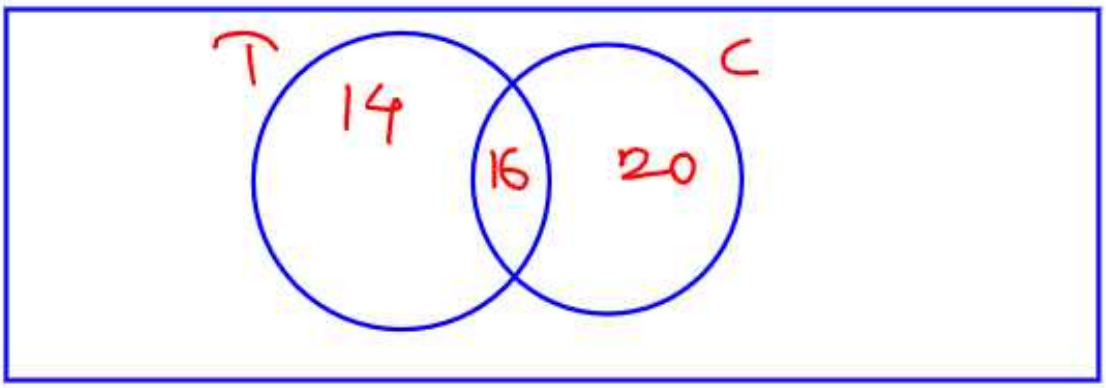


Let's Solve

Ex. In a group of 50 persons, 14 drink tea but not coffee and 30 drink tea. Find

(i) How many drink tea and coffee both? 16

(ii) How many drink coffee but not tea? 20



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Let's Solve

Ex. In a group of 50 persons, 14 drink tea but not coffee and 30 drink tea. Find

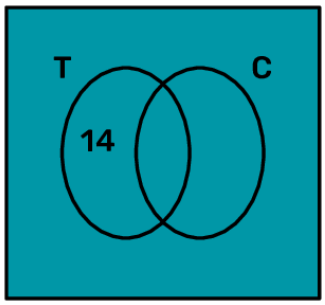
(i) How many drink tea and coffee both?

(ii) How many drink coffee but not tea?

Solution

T : People drinking tea

C : People drinking coffee



(i) $n(T) = n(T - C) + n(T \cap C)$

$\Rightarrow 30 = 14 + n(T \cap C)$

$\Rightarrow n(T \cap C) = 16$

(ii) $n(C - T) = n(T \cup C) - n(T) = 50 - 30 = 20$

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Let's Solve

Ex. In a city 20 percent of the population travels by car, 50 percent travels by bus and 10 percent travels by both car and bus. Then persons travelling by car or bus is

A 80%

B 40%

C 70%

D 60%

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Let's Solve

Ex. In a city 20 percent of the population travels by car, 50 percent travels by bus and 10 percent travels by both car and bus. Then persons travelling by car or bus is

A 80% B 40% C 70% D 60%

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Let's Solve

Ex. In a city 20 percent of the population travels by car, 50 percent travels by bus and 10 percent travels by both car and bus. Then persons travelling by car or bus is

Solution

Let total population of city = 100

$n(C) = 20$

$n(B) = 50$

$n(B \cap C) = 10$

$$n(B \cup C) = n(B) + n(C) - n(B \cap C) = 20 + 50 - 10$$
$$= 60$$

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Let's Solve



Let's Solve

Ex. In group of 500 people, 350 speak Hindi and 300 speak English. It is given that person speaks at least one language.
(i) How many people can speak both Hindi and English?
(ii) Ho many people can speak Hindi only?
(iii) How many people can speak English only?

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Let's Solve

Ex. In group of 500 people, 350 speak Hindi and 300 speak English. It is given that person speaks at least one language.
(i) How many people can speak both Hindi and English?
(ii) Ho many people can speak Hindi only?
(iii) How many people can speak English only?

Solution

Let H be the set of people who speak Hindi, and E be the set of people who speak English.

$\therefore\ n(H \cup E) = 500,\ n(H) = 350,\ n(E) = 300$

(i) We have to find $n(H \cap E)$

Now, $n(H \cup E) = n(H) + n(E) - n(H \cap E)$

$\therefore\ 500 = 350 + 300 - n(H \cap E)$

$\Rightarrow 500 = 650 - n(H \cap E)$

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Let's Solve

Solution

$\Rightarrow n(H \cap E) = 650 - 500$

$\therefore\ n(H \cap E) = 150$

Thus, 150 people can speak both Hindi and English

(ii) Number of people who speak Hindi only $= n(H) - n(H \cap E)$

$= 350 - 150 = 200$

(iii) Number of people who speak English only $= n(E) - n(H \cap E)$

$= 300 - 150 = 150$



Did you understand everything till now? Poll

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Quiz.

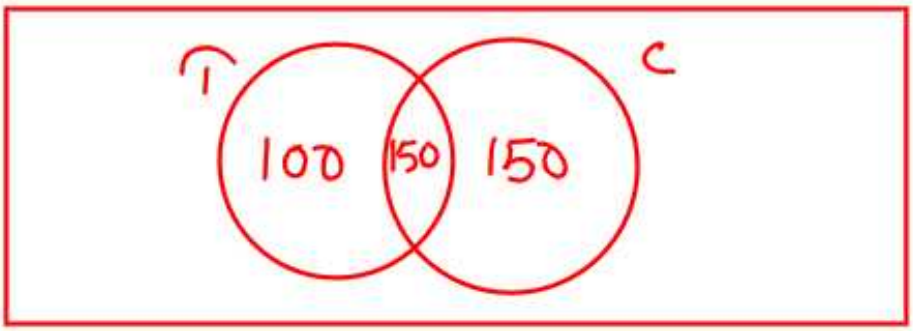


Quiz.

Chat Off

Q. In a survey conducted on 800 students of a school, 250 students were found to like tea and 300 like coffee, 150 like both tea and coffee. Find how many students like neither tea nor coffee.

A 350 B 400 C 150 D 450



100+150+150

400

800 − 400 = 400

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Quiz.

Chat On

Q. In a survey conducted on 800 students of a school, 250 students were found to like tea and 300 like coffee, 150 like both tea and coffee. Find how many students like neither tea nor coffee.

A 350 B 400 C 150 D 450



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Quiz.

Q. In a survey conducted on 800 students of a school, 250 students were found to like tea and 300 like coffee, 150 like both tea and coffee. Find how many students like neither tea nor coffee.

Solution

Let U be the set of all students who took part in survey
T be the set of students who like tea,
C be the set of students who like coffee.

$$n(U)=800,\ n(T)=250\ \ n(C)=300,\ n(T\cap C)=150$$

$$n(T'\cap C')=n\left((T\cup C)'\right)$$

$$=n(U)-n(T\cup C)$$

$$=n(U)-\left[n(T)+n(C)-n(T\cap C)\right]$$

$$=800-\left[250+300-150\right]=400$$

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Let's Solve



Let's Solve

Ex. In statistical survey of 1003 families of Kolkata, it was found that 63 families have neither a radio nor a TV, 794 families have a radio and 187 have TV. The number of families in that group having both a radio and a TV is

A 36 **B** 41 **C** 32 **D** None

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Let's Solve

Ex. In statistical survey of 1003 families of Kolkata, it was found that 63 families have neither a radio nor a TV, 794 families have a radio and 187 have TV. The number of families in that group having both a radio and a TV is

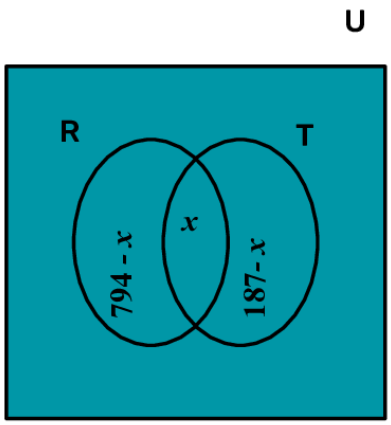
A 36 B 41 C 32 D None

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Let's Solve

Ex. In statistical survey of 1003 families of Kolkata, it was found that 63 families have neither a radio nor a TV, 794 families have a radio and 187 have TV. The number of families in that group having both a radio and a TV is

Solution



Let R be the set of families having a radio and T be the set of families having a TV.

Then,

$n(R \cup T)$ = Number of families having at least one of radio and TV

$= 1003 - 63 = 940$

$n(R) = 794$ and $n(T) = 187$

Let x be the number of families having both radio and a TV

Then number of families who have only Radio $= 794 - x$

Number of families who have only TV $= 187 - x$

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Let's Solve

Ex. In statistical survey of 1003 families of Kolkata, it was found that 63 families have neither a radio nor a TV, 794 families have a radio and 187 have TV. The number of families in that group having both a radio and a TV is

Solution

From venn diagram

$794 - x + x + 187 - x = 940$

$\Rightarrow 981 - x = 940$

$\therefore\ x = 981 - 940 = 41$

Hence, the required number of families having both radio and TV is 41

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1.6 Cardinality based problems



1.6 Cardinality based problems

Result

(i) $n(A \cup B \cup C) = n(A) + n(B) + n(C)$

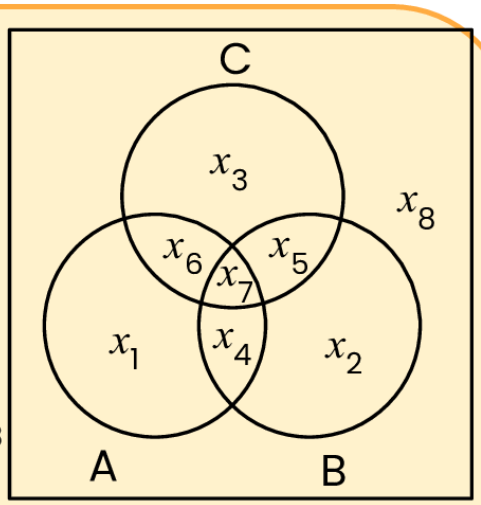
$$- n(A \cap B) - n(A \cap C) - n(B \cap C)$$

$$+ n(A \cap B \cap C)$$

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1.6 Cardinality based problems

Remark



(a) n(only one of A, B or C) = $x_1 + x_2 + x_3$

(b) n(exactly two of A, B and C) = $x_4 + x_5 + x_6$

(c) $n(A, B \text{ or } C) = x_1 + x_2 + x_3 + x_4 + x_5 + x_6 + x_7$

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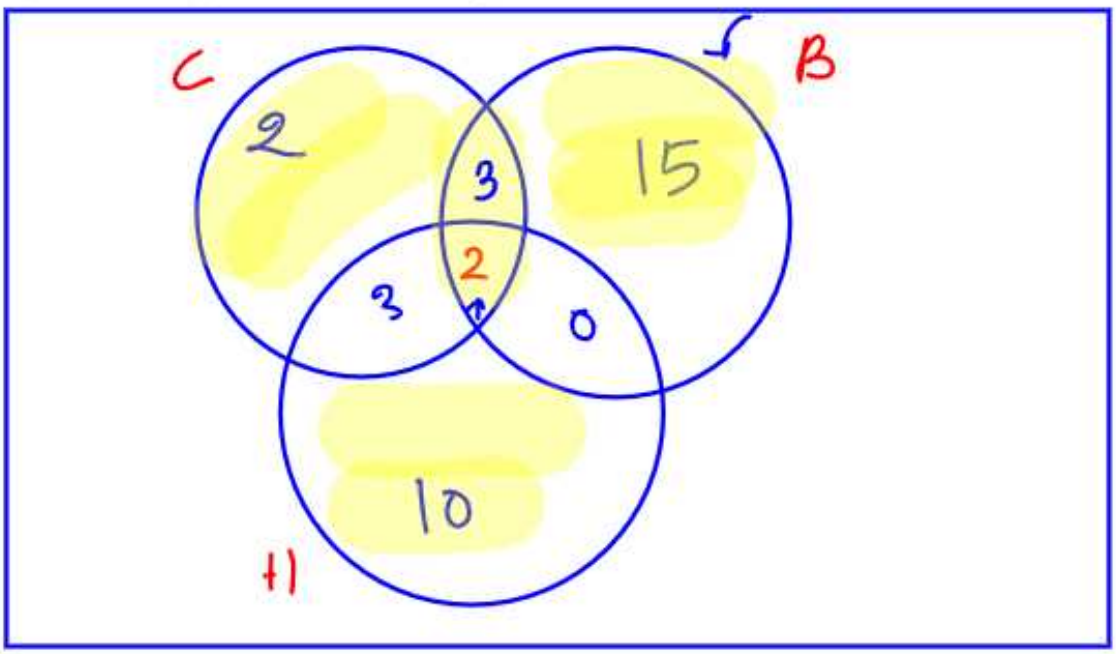
Let's Solve



Let's Solve

$2+15+10=27$

Ex. In a class, every student plays either cricket or hockey or badminton. 10 play cricket, 15 play hockey, 20 play badminton, 5 play cricket and hockey, 2 play hockey and badminton, 5 play badminton and cricket and 2 play all the games. How many play exactly one of the games?



Let's Solve

Ex. In a class, every student plays either cricket or hockey or badminton. 10 play cricket, 15 play hockey, 20 play badminton, 5 play cricket and hockey, 2 play hockey and badminton, 5 play badminton and cricket and 2 play all the games. How many play exactly one of the games?

Solution

$$n(C) = 10 = x_1 + x_4 + x_6 + x_7$$

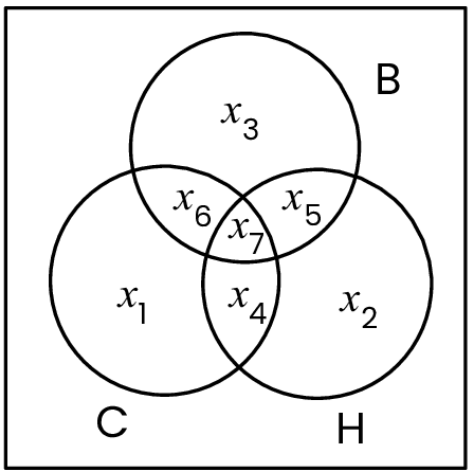
$$n(H) = 15 = x_2 + x_4 + x_7 + x_5$$

$$n(B) = 20 = x_3 + x_5 + x_6 + x_7$$

$$n(C \cap H) = 5 = x_4 + x_7$$

$$n(H \cap B) = 2 = x_5 + x_7$$

$$n(B \cap C) = 5 = x_6 + x_7$$



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Let's Solve

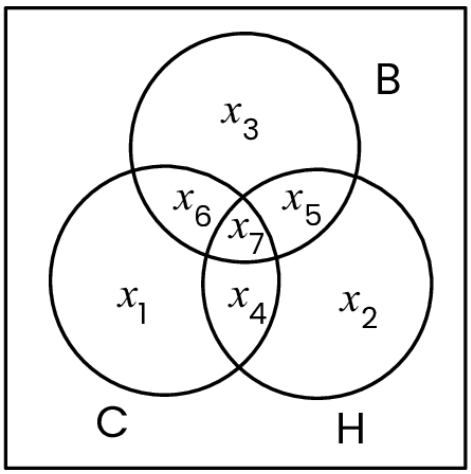
Ex. In a class, every student plays either cricket or hockey or badminton. 10 play cricket, 15 play hockey, 20 play badminton, 5 play cricket and hockey, 2 play hockey and badminton, 5 play badminton and cricket and 2 play all the games. How many play exactly one of the games?

Solution

$$n(C \cap H \cap B) = 2 = x_7$$

$$\therefore x_4 = 3;\ x_5 = 0;\ x_6 = 3$$

$$x_1 = 2;\ x_2 = 10;\ x_3 = 15$$



Number of students playing exactly one game

$= x_1 + x_2 + x_3 = 27$

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Let's Solve

Ex. Out of 100 students; 15 passed in English, 12 in Mathematics, 8 in Science, 6 in English and Mathematics, 7 in Mathematics and Science, 4 in English and Science and 4 in all the three. Find how many students passed

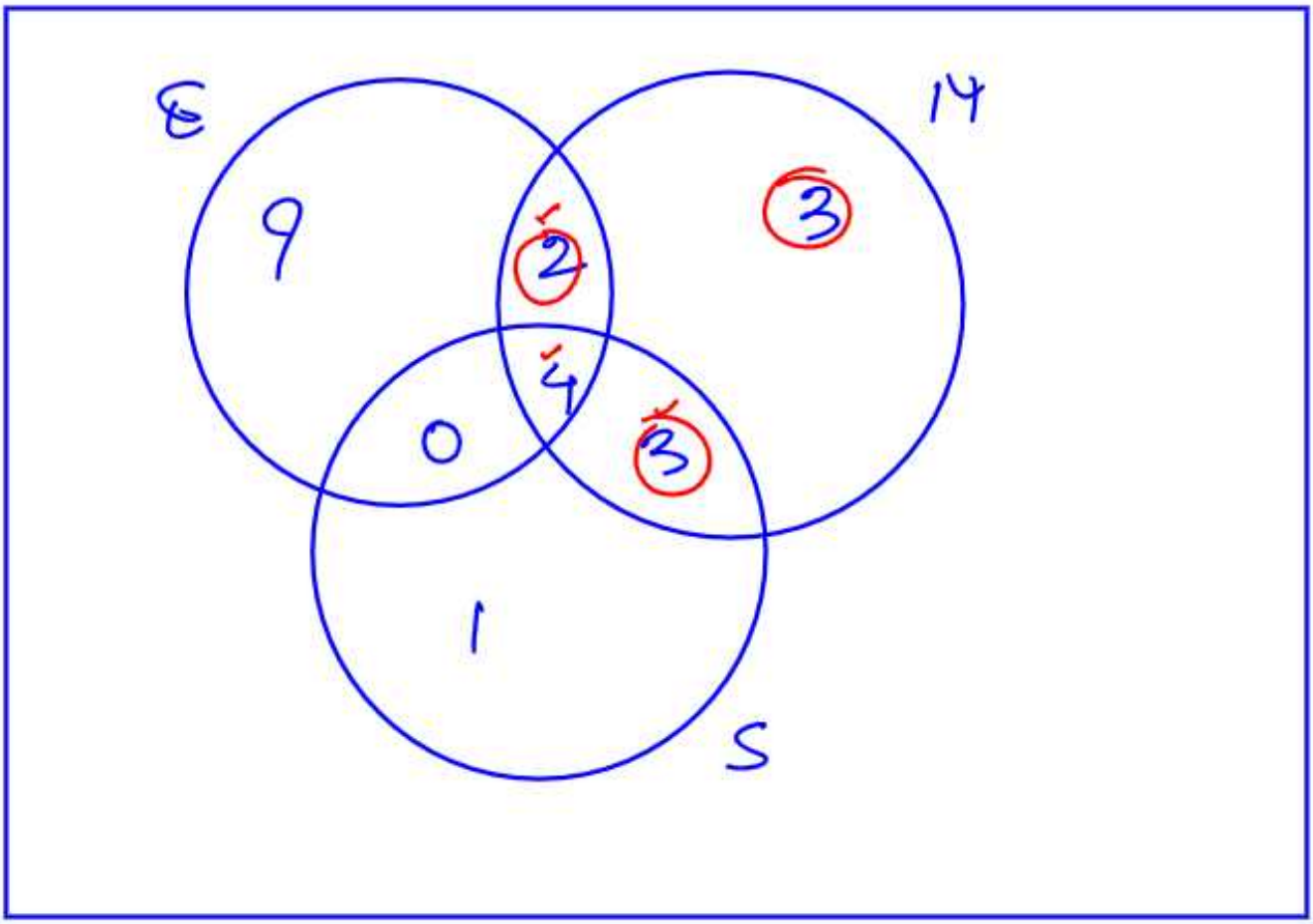
(i) in English and Mathematics but not in Science

(ii) in Mathematics and Science but not in English

(iii) in Mathematics only

(iv) in more than one subjects

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ε
9
14
3
2
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1
S

Let's Solve

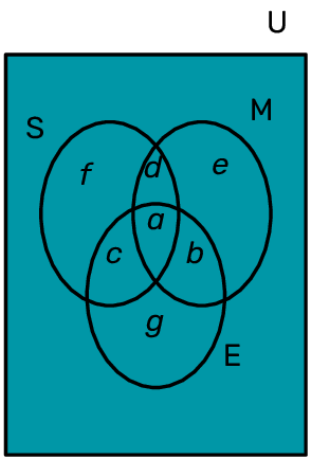
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(i) in English and Mathematics but not in Science

(ii) in Mathematics and Science but not in English

(iii) in Mathematics only

(iv) in more than one subjects



Solution

Let M be the set of students who passes in Mathematics, E be the set of students who passes in English and S be the set of students who passed in Science

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Solution

Given,

$n(U) = 100,$

$n(E) = 15,\ n(M) = 12,\ n(S) = 8,$

$n(E \cap M) = 6,\ n(M \cap S) = 7,$

$n(E \cap S) = 4$ and $n(E \cap M \cap S) = 4,$

From the diagram, we have

$a = n(E \cap M \cap S) = 4$ and $a + d = n(M \cap S) = 7 \Rightarrow d = 3$

$a + b = n(M \cap E) = 6 \Rightarrow b = 2$

$a + c = n(S \cap E) = 4 \Rightarrow c = 0$

$a + b + d + e = n(M) = 12$

$\Rightarrow 4 + 2 + 3 + e = 12 \Rightarrow e = 3$

$a + b + c + g = n(E) = 15$

$\Rightarrow 4 + 2 + 0 + g = 15 \Rightarrow g = 9$

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Let's Solve

Solution

$a + c + d + f = n(S) = 8$

$\Rightarrow 4 + 0 + 3 + f = 8 \Rightarrow f = 1$

Number of students passed in

(a) English and Mathematics but not in Science $= b = 2$

(b) Mathematics and Science but not in English $= d = 3$

(c) Mathematics only $= e = 3$

(d) more than one subject $= a + b + c + d = 4 + 2 + 0 + 3 = 9$

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Quiz.



Quiz.

Chat Off

Q. In a town of 10,000 families, it was found that 40% families buy newspaper A, 20% buy newspaper B and 10% buy newspaper C. Also, 5% families buy newspapers A and B, 3% buy newspapers B and C and 4% buy newspapers A and C. If 2% families buy all the three newspapers, then number of families which buy newspaper A only is

A 3100 B 3300 C 2900 D 1400



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Solution

$n(A) = 40\%$ of $10,000 = 4,000$

$n(B) = 20\%$ of $10,000 = 2,000$

$n(C) = 10\%$ of $10,000 = 1,000$

$n(A \cap B) = 5\%$ of $10,000 = 500$

$n(B \cap C) = 3\%$ of $10,000 = 300$

$n(C \cap A) = 4\%$ of $10,000 = 400$

$n(A \cap B \cap C) = 2\%$ of $10,000 = 200$

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Quiz.

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Solution

Number of families which buy newspaper A only

$$= n(A) - \left[n(A \cap B) + n(A \cap C) - n(A \cap B \cap C)\right]$$

$$= 4000 - \left[500 + 400 - 200\right]$$

$$= 4000 - 700$$

$$= 3300$$

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Quiz.



Quiz.

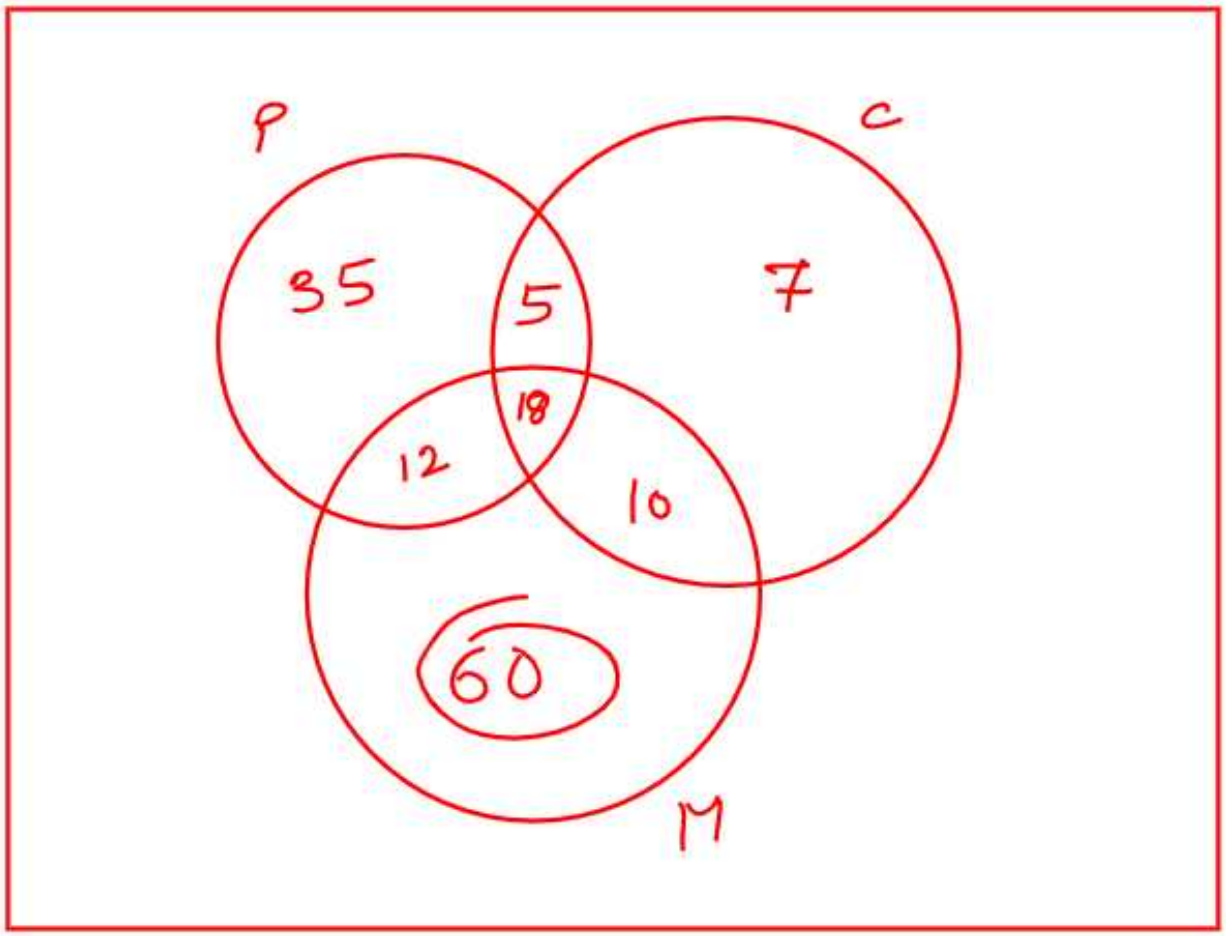
Chat Off

Q. A class has 175 students. The following data shows the number of students studying different subjects. Mathematics 100, Physics 70, Chemistry 40, Mathematics and Physics 30, Mathematics and Chemistry 28, Physics and Chemistry 23, Mathematics & Physics & Chemistry 18. How many students study Mathematics alone?

A 35 **B** 48 **C** 60 **D** 22



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P
C
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7
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60
M

Quiz.

Chat On

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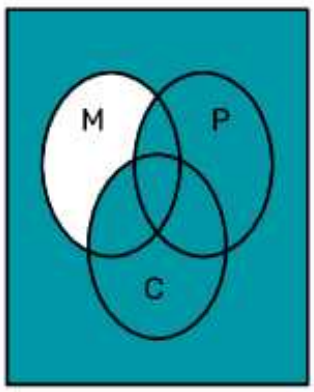
Solution

n(M alone)

$$= n(M) - n(M \cap C) - n(M \cap P) + n(M \cap C \cap P)$$

$$= 100 - 28 - 30 + 18$$

$$= 60$$



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